Data Warehouse (by Ansh Lamba)

Q) What is Data warehouse?
* Central location to store every single data.

Q) What is Data warehousing?
* Process that you follow to fetch data from Database & store it to Data warehouse
* By using ETL we transfer data from database to DWH

* DWH holds or stores a whole data of a organization.
* Data mart is a subpart (subset) of DWH it might be finance domain part, salary domain part like that.

Incremental loading :-
* By using CDC (Change Data Capture) we can achive Incremental loading.

Q) What is data modeling?
* Divide one big table to small small tables.

* By using fact table & Dimentation table we can achive data modeling.

* Fact table consist of Numarical Values & Foreign keys of dimention table.
* Dimention table Consist of set of related data & foreign key to communicate with Fact table to use numarical values.

Big table

order Id	order_name	Product ID	Product_Name	Quanity	revanue

* Above table divides into Dimention tables first

Order Id	order_name	foreign_key

This is one dimention table

Product ID	Product_name	f_k

This is another dimention table.

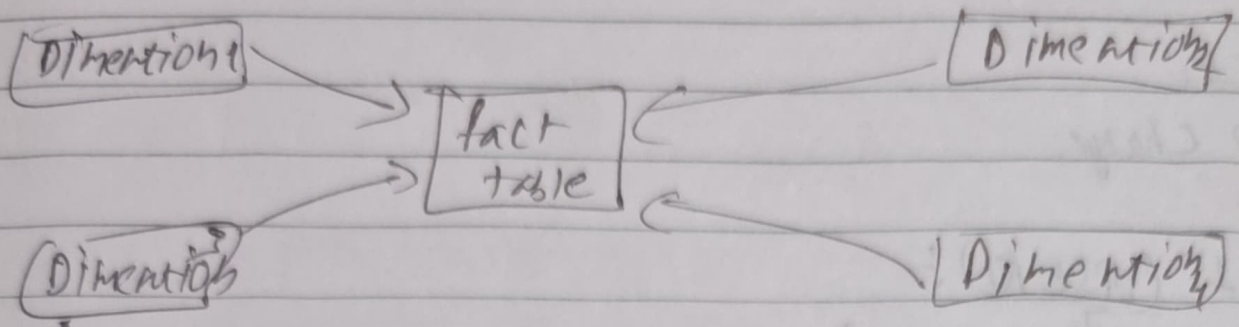
* By using row_number window function we have to create a foreign key.

Quanity	revanue	F_K_Order	F_K_Product

This is fact table.

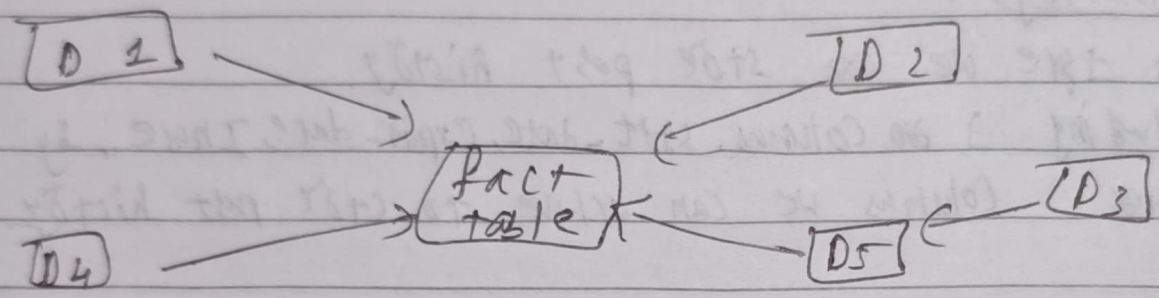
* By using these two fact table & dimention tables we can create a schemas like star schema, snowflake schema & galaxy schema.
* 90% of industry use star schema.

Star schema:



* All Dimention ~~the~~ tables directly use fact table.

Snowflake schema:



* It allows indirect communication so it is complex that's why most of the industries uses star schema.
* By using Join operation to D3 & D5 they will make snow schema to star schema.

SLOWLY Changing dimensions:-

* we mainly use 4 types → Type0, Type1, Type 2, Type3.
* It useful for future business purpose if we want to launch new product or rename existing product at that time it is useful rather than building from scratch.
* Purpose for this is modify existing fact table or add new columns to fact table.

Type 0:-

* No Change.

Type 1 (UPSERT) :-

(Update + Insert).

* After update is doesn't store past history.

Type 2 (History) :-

* In this type we can store past history.
* By including 3 Columns. start_date, expair_date, Image, by using these 3 Columns we can achive to store past history.

Type 3 (Previous History only) :-

* It stores only previous history value.

Ex for Type 1:-

Merge into Table_name_full_path as allias
Using Table_name As allias
ON Column_name = Column_name
when matched then update SET *
when not matched then insert *.